

HARUN ENSARIOĞLU

Robotic Software Developer

+90 543 774 0868

ensariharun4934@gmail.com

Istanbul, Türkiye

github.com/HarunEnsar

/in/harun-ensarioğlu

SUMMARY

I am a Computer Engineer with expertise in **artificial intelligence, computer vision, UAVs, and robotics (quadruped and humanoid systems)**. I'm disciplined, punctual, and a fast learner who values teamwork and continuous skill development. I love research, quickly explore new technologies, and regularly review **academic papers and projects**. I have actively participated in field-related projects and approach my work with attention to detail, aiming to make meaningful contributions in the technology sector through collaborative innovation.

SKILLS

Languages: Python, C++, C, C#, SQL.

Core: Reinforcement Learning, LeRobot, PID, Image Processing,

Tools: ROS, Linux, Gazebo, MoveIt, NVIDIA Isaac (Sim/Lab/Gym), MAT-LAB/Simulink

PROJECTS

- 2209-A TUBITAK **Performance Comparison of NMPC and PID Control Systems for Directional Stability in UAV Systems**
This study compares PID and NMPC control methods for directional stability in UAVs. Using a quadrotor UAV model, simulations and real-time experiments demonstrate that NMPC outperforms PID in stability, response time, and robustness to disturbances.
- Bachelor's Thesis **Moving Platform UAV Landing** https://github.com/HarunEnsar/moving_platform_uav_landing
Low-cost UAV landing system using ArUco markers was developed with Ubuntu, ROS, Gazebo, and ArduPilot SITL. Python-based algorithms enable precise landings on moving platforms. This system offers a practical, accurate, and efficient solution for military and civil use.
- TEKNOFEST **UAV and Computer Vision competitions**
Participation in TEKNOFEST-organized competitions: Artificial Intelligence in Transportation, International UAV, and Agricultural Technologies
- Mobile App **Explore Türkiye Android App** https://github.com/HarunEnsar/Explore_Turkiye_Android_App
A Flutter application that showcases the must-visit destinations in Turkey.

EDUCATION

- 9/2021 - 7/2025 **Computer Engineering - ISTANBUL GEDİK UNIVERSITY** **University**
GPA: 3.68 : Ranked 2nd in Department and 3rd in Faculty with 3.68 GPA
Activities and Societies: Arkhe UAV Team Captain. Advisory Board Member, Departments of Computer and Software Engineering
- 9/2020 - 9/2021 **Electrical and Electronics Engineering - MERSIN UNIVERSITY** **University**
English preparatory

EXPERIENCE

- 2/2024 - **Robotic Software Developer** **LTC Innovation**
- Hardware and software integration for AI and deep learning projects using NVIDIA Jetson
 - Conducting and deploying Reinforcement Learning trainings using Isaac Lab and Isaac Gym
 - Conducting and deploying Imitation Learning trainings with humanoid robots
 - Developing and optimizing XR-based teleoperation software for whole body/upper body manipulation of humanoid robots
 - Developing robot control software and sensor data processing using ROS (Robot Operating System)
 - Conducting autonomous robot simulations using Isaac Sim (humanoid, quadruped, and mobile robots)
 - Developing Speech-to-Text (converting voice commands to text) and Text-to-Speech (converting text to speech) applications (providing AI-supported verbal interaction for robotic dogs)
 - Robot simulation and control using Gazebo, MuJoCo, and ROS
- NVIDIA Jetson, ROS, Isaac Sim, Isaac Lab, Isaac Gym, Gazebo, MuJoCo, Python, TensorFlow/PyTorch, Speech-to-Text, Text-to-Speech

2/2024 – 8/2024	Long Term Software Intern • Simulation of UAVs in a three-dimensional environment using DroneKit and Gazebo, incorporating image processing and GPS-based navigation • Image transmission utilizing MAVLink and telemetry protocols • Detection of diseased plants via UAV-captured imagery • Development of a Python API for interpreting UAV-acquired images through artificial intelligence-based image processing • Design and development of a drone management platform using ASP.NET Core MVC and React JS • Android application development using Python DroneKit, Gazebo, MAVLink, Python, OpenCV, TensorFlow, ASP.NET Core MVC, React JS, SQL	Mostas Technology
7/2023 – 9/2023	Student Recruitment Specialist Corporate Communication and Promotion Department Presidency	Istanbul Gedik University

LANGUAGES

English - B1